Rush Hour: Bike-Sharing Demand - Project 1 report

Machine Learning, Winter 2026 | Team IDs 16002932, 16004208, 16006616 | team seed 42648. All numbers and plots come from **bike_sharing_pipeline.ipynb** (one notebook, five chained phases plus the test predictions, run top to bottom).

Terms. R^2 = share of the variation in demand that the model explains (1 is perfect). RMSE = typical error in bikes per hour. Underfitting = the model is too simple. Overfitting = it memorises the training data. Chronological split = train on the early dates, test on the latest dates. Numbers are on the 20% validation set unless we say otherwise.

Setup used in every phase

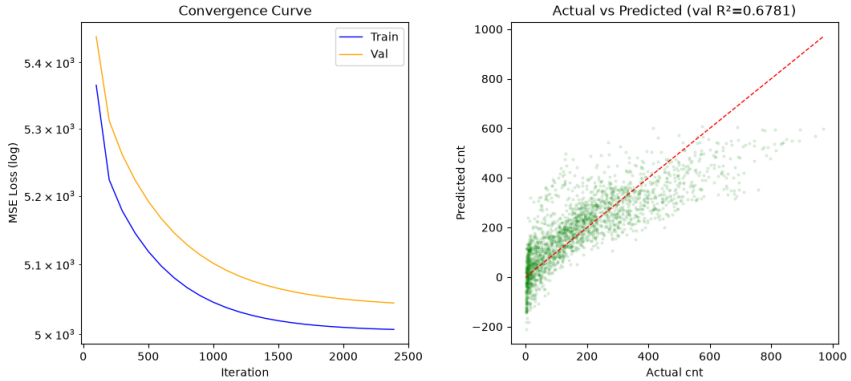
- **Data:** only *train.csv* (10,886 hourly rows, days 1-19 of every month), *test.csv* (574 rows, the 20th of every month, no labels) and *sample_submission.csv*.
- **Seed:** sha256 of the sorted IDs joined by '_', modulo 100,000 = **42648** (the PDF's example IDs give 41698 in our code). One seeded 80/20 split: 8,708 training and 2,178 validation rows.
- **No leakage:** scalers, encoders, polynomial terms and feature selection are fitted on training rows only; *test.csv* is only transformed.
- **Chaining:** each phase keeps its weights and settings in memory with the SHA-256 of the weights; the next phase checks that hash before it starts from them.
- **No sklearn model is fitted.** sklearn is used only for the split, the day-grouped folds and metric cross-checks; a cell in the notebook scans every executed cell for fit/predict calls.
- **Speed:** Phases 3-5 and the final fit reuse the Phase 1 gradient descent with an optional momentum step (Nesterov, restarted when the loss rises). Plain gradient descent needed up to 150,000 steps on the 589-feature models; the notebook now runs in minutes.

Phase 1 - Gradient descent

In one sentence: a straight-line model trained with our own gradient descent reaches validation R^2 0.678 (RMSE 100.44): it works, but it is too simple.

choice	what we picked	why
features	one-hot hr, season, mnth, weekday, weathersit; dteday becomes days_since_start; 5 numeric columns standardised; a flag for windspeed = 0 (54 features)	demand peaks at 8 a.m. and 5 p.m., so hr cannot be one number; windspeed is exactly 0 in 12.1% of rows (looks like a missing-value code); weathersit = 4 occurs once, merged into 3
learning rate	0.619 (= 1.5/L with L = 2.4219)	gradient descent is stable only below 2/L = 0.826. Sweep: up to 0.1 too slow, 1/L to 1.9/L converge, 2/L oscillates (loss 25,620), 2.1/L diverges. 1.5/L is fast and safe
stopping rule	loss change below 1e-6 and gradient norm below 1	stops when the loss is flat; converged after 2,389 iterations

- Train R^2 0.698 against validation 0.678 (gap 0.02): the model **underfits**, it does not overfit.
- It matches closed-form least squares: training loss differs by 1.7e-6, validation R^2 0.678543 vs 0.678527. Gradient check passes (error 1.6e-6).
- The matrix has rank 50 of 54 (season repeats the month; workingday follows weekday and holiday), so single weights are not trustworthy, but predictions are.
- 9.55% of predictions are negative (clipping at 0 would give R^2 0.6966); we keep them in the chain and floor only the final test predictions.
- **Bonus (under-predicting costs 3 times more):** with e = y - ŷ and c = 3 if e > 0 else 1, J = mean(c·e²), so ∂J/∂w = -(2/n)·Xᵀ(c·e) and ∂J/∂b = -(2/n)·Σc·e; the step is 1/(6L) = 0.0688. Under-predicted hours fall from 46.69% to 32.69%, the mean residual moves from -1.80 to -43.36, the asymmetric cost drops from 20,602 to 17,448, and R^2 falls from 0.6781 to 0.5961.



Phase 1: convergence of the final fit (left) and actual versus predicted validation counts (right).

Phase 2 - Polynomial regression

In one sentence: adding powers of the continuous columns, started from the Phase 1 weights, only lifts validation R^2 from 0.678 to 0.686; curves are not what is missing.

choice	what we picked	why
columns to expand	temp, hum, windspeed, days_since_start	powers of 0/1 dummies add nothing; atemp is almost temp (r = 0.985), adding its powers lowers R^2 from 0.6860 to 0.6854
starting point	Phase 1 weights, zeros for the new terms	predictions at step 0 equal Phase 1's (difference 2e-13); at degree 2 it converges in 178 iterations instead of 3,603 from zero
degree	3 (62 features)	smallest degree within 0.001 of the best usable one (degree 4: 0.6865). Degrees 5-6 are flagged hard for gradient descent (condition number 1.3e4 and 2e5)

- Validation R^2 for degrees 1 to 6: 0.6781, 0.6796, 0.6858, 0.6865, 0.6864, 0.6866. The chosen model has RMSE 99.24 (+0.0077 over Phase 1); the train-validation gap stays about 0.02.
- Evidence of curvature is weak: single-feature quadratics change R^2 by -0.001 to +0.004.
- The Phase 1 mistakes are structured by hour and day type (mean residual -277 at 8 a.m. on non-working days, +89 on working days), which no weather power can fix: our hypothesis for Phase 3. Cross terms between the four continuous columns reach 0.7146 (closed form).

Phase 3 - Bias-variance tradeoff

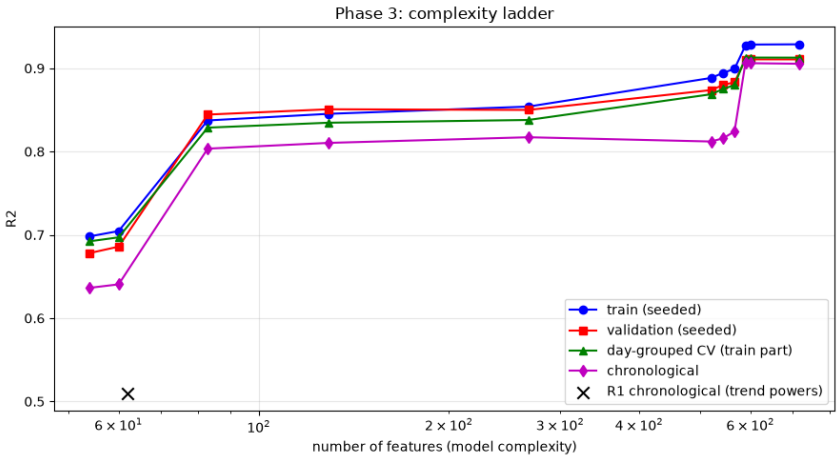
In one sentence: the problem is underfitting, not overfitting: hour interactions lift validation R^2 from 0.686 to 0.910, while powers of the time index break predictions on later dates.

choice	what we picked	why
what to vary	a ladder of 12 nested models: R0 linear, R1 Phase 2's model, R2 weather powers only, then one hour-interaction family per rung (R3-R11)	Phase 2 showed powers do not help, so the missing piece must be interactions
how to judge	seeded validation, day-grouped 5-fold CV inside the training part, and a chronological split	CV keeps whole days together and does not touch the validation set; the chronological split tests new dates
chronological cut	sorted by date and hour, first 80% train (cut after 2012-08-05)	the PDF asks for earlier-versus-later; this split uses no seed
diagnosis rule (fixed before running)	bias: gap < 0.02 and CV still rising by > 0.005; variance: gap > 0.04 or CV/chronological falling	set in advance so the diagnosis is not chosen after seeing the numbers
which model to carry on	least complex rung within 0.005 of the best CV R^2 (0.9128, threshold 0.9078): R9, 589 features	if two models are equally good, keep the simpler one

rung	features	train R^2	validation R^2	gap	CV R^2	CV std	chronological R^2
R0 linear (Phase 1)	54	0.6982	0.6781	0.0200	0.6923	0.0132	0.6364
R1 Phase 2 model (+ days powers)	62	0.7044	0.6858	0.0186	0.6969	0.0103	0.5099
R2 weather powers only	60	0.7045	0.6859	0.0187	0.6970	0.0103	0.6406
R3 + hr x workingday	83	0.8373	0.8444	-0.0070	0.8287	0.0101	0.8035
R4 + hr x weather	129	0.8453	0.8507	-0.0055	0.8346	0.0069	0.8104
R5 + hr x weekday	267	0.8540	0.8499	0.0040	0.8379	0.0080	0.8172
R6 + hr x mnth	520	0.8884	0.8738	0.0146	0.8688	0.0060	0.8120
R7 + hr x temp	543	0.8945	0.8799	0.0145	0.8752	0.0082	0.8159
R8 + hr x hum	566	0.8992	0.8832	0.0160	0.8797	0.0087	0.8241
R9 + hr x yr (carried forward)	**589**	**0.9273**	**0.9104**	**0.0170**	**0.9125**	**0.0058**	**0.9063**
R10 + workingday x yr, x mnth	601	0.9284	0.9107	0.0178	0.9128	0.0054	0.9061
R11 + hr x windspeed, season, holiday	716	0.9287	0.9106	0.0181	0.9127	0.0053	0.9054

- **Underfitting:** the largest train-validation gap is 0.020 (0.017 at R9), far below the 0.04 overfitting line. Capacity keeps paying (CV +0.132 at R3, +0.031 at R6, +0.033 at R9), then flattens after R9 (+0.0003, -0.0001).
- **The time index is the trap:** removing the powers of days_since_start (R1 to R2) changes validation by -0.0001 and CV by +0.0001, but the chronological R^2 jumps from 0.5099 to 0.6406. At R9, keeping them gives 0.8508 instead of 0.9063. Growth over time is carried by hr x yr instead.
- **Carried forward (R9, 589 features):** train 0.9273, validation 0.9104 (RMSE 53.00), CV 0.9125 ± 0.0058, chronological 0.9063, holdout of days 18-19 0.8967. The learning-curve gap is +0.4156 with 10% of the data and +0.0170 with all of it.

- **Which estimate to trust:** the random split hides the time problem (Phase 2's model: 0.6858 on validation but 0.5099 chronologically), so for new periods read the chronological number. *test.csv* is the 20th of every month, inside the training period, so the CV and the days 18-19 holdout are its closest proxies. All four estimates lie between 0.897 and 0.913.



Phase 3: R^2 along the ladder (black cross: Phase 2's model on the chronological split).

Phase 4 - Regularization (L1, L2, Elastic Net)

In one sentence: all three methods match the unregularised model (about 0.907-0.908 against 0.910); Lasso is the only one that removes features, so we carry it on with 470 of 589 features.

choice	what we picked	why
methods	Ridge (L2) with our gradient descent; Lasso (L1) and Elastic Net with proximal gradient descent (soft-thresholding)	Elastic Net mixes L1 and L2 with a share α (0.25, 0.5, 0.9)
λ grid	1e-4 to 10, 26 values (16 for Elastic Net), warm start from the larger λ	Phase 3 said underfitting, so we expected small λ to win: all methods peak at $\lambda \leq 0.07$
step size	0.9/(L + λ(1-α))	the L2 term adds curvature; the plain 0.9/L step diverges for λ above about 4
choosing λ	largest λ within 0.005 of that method's best validation R^2	the same rule for all three: the simplest model that is still equally good
winner	Lasso, λ = 0.2512	accuracy is equal within the CV spread (±0.006); only Lasso removes features

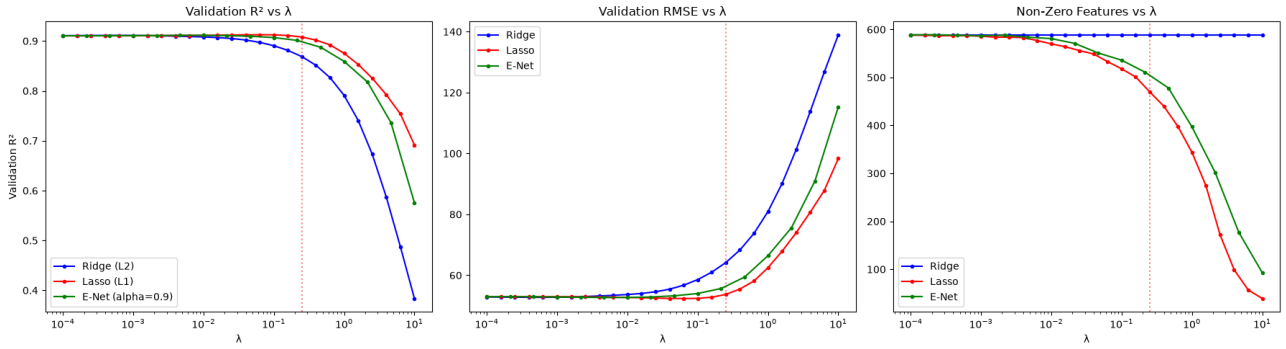
method	best R^2 (at λ)	chosen λ	validation R^2	RMSE	features kept
Ridge (L2)	0.9112 (0.0006)	0.01585	0.9069	54.03	589 of 589
Lasso (L1)	0.9125 (0.063)	0.2512	0.9079	53.72	**470** of 589
Elastic Net (α = 0.9)	0.9114 (0.0046)	0.1	0.9069	54.02	536 of 589
unregularised (Phase 3)	-	0	0.9104	53.00	589 of 589

Verdict for every original column. We removed all features derived from one column (its dummies, powers and interaction terms) and refitted the chosen Lasso. **Useful** if R^2 drops by 0.001 or more; otherwise **Redundant** if $|r| \geq 0.85$ with another column; otherwise **Uninformative**.

column	kept / derived	ΔR²	closest partner	verdict
hr	455 / 552	0.5752	hum (0.28)	Useful
yr	22 / 24	0.0306	instant (0.87)	Useful
workingday	22 / 24	0.0190	holiday (0.25)	Useful
mnth	230 / 264	0.0113	season (0.97)	Useful
weathersit	38 / 48	0.0103	hum (0.40)	Useful
temp	24 / 26	0.0079	atemp (0.98)	Useful
hum	22 / 26	0.0052	weathersit (0.40)	Useful

column	kept / derived	ΔR²	closest partner	verdict
weekday	100 / 144	0.0029	holiday (0.11)	Useful
instant	0 / 0	0.0000	dteday (1.00)	Redundant
dteday	1 / 1	0.0004	instant (1.00)	Redundant
season	1 / 3	-0.0000	mnth (0.97)	Redundant
atemp	1 / 1	0.0002	temp (0.98)	Redundant
holiday	0 / 1	-0.0000	workingday (0.25)	Uninformative
windspeed	2 / 4	0.0005	hum (0.32)	Uninformative

- hr is by far the strongest column. instant and dteday are perfectly correlated (instant is only a counter), season repeats mnth, atemp repeats temp. holiday has all its terms zeroed (workingday already encodes it); windspeed adds only 0.0005.
- yr has $|r| = 0.87$ with instant but is useful, because hr x yr carries the growth over time; this is why the rule looks at the R^2 drop first.
- **Chronological check:** the chosen Lasso scores 0.8938 (RMSE 70.91) against 0.9063 for the unregularised model: pruning costs 0.0025 on the seeded split and 0.0125 on later dates.



Phase 4: validation R², RMSE and number of non-zero features against λ (dotted line: the chosen Lasso λ).

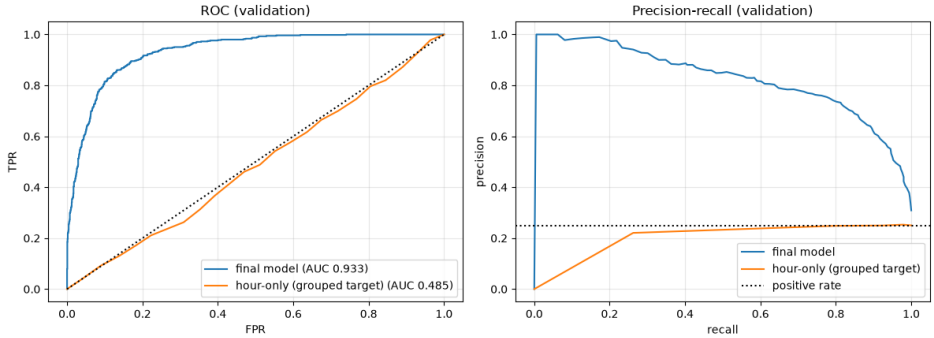
Phase 5 - Logistic regression

In one sentence: a logistic model on the 470 features Phase 4 kept predicts whether an hour is busy for its kind of hour, with ROC-AUC 0.933 and F1 0.770 on validation.

choice	what we picked	why
what is 'high demand'	cnt above the training 75th percentile of the same hour and day type (48 groups, computed on training rows only)	one global threshold (283 bikes) calls every rush hour 'high': 63.8% of its positives fall in 7 rush hours (an even spread would be 29.2%), and a model that only knows the hour gets AUC 0.850 on it. On our definition the same model gets 0.485 (guessing)
features	the 470 features Phase 4 kept (hash-checked)	the PDF says to use only the features that survived Phase 4
L2 and class weights	L2 = 0.01, no class weights	12 settings tried (6 L2 values x weights off/on); AUC 0.883 to 0.936; 4 settings are within 0.005 F1 of the best, we take the most regularised. Class weights only shift the best threshold
metrics	F1 and ROC-AUC, plus accuracy	75% of the hours are 'not high', so 'always no' already scores accuracy 0.750
decision threshold	0.38 (best F1 on validation)	tuned on the same validation set, so the tuned numbers are slightly optimistic

model	accuracy	precision	recall	F1	ROC-AUC
logistic, 470 features, threshold 0.38	0.8788	0.7322	0.8128	**0.7704**	**0.9331**
same model, threshold 0.5	0.8747	0.7845	0.6881	0.7331	0.9331
baseline: always 'not high'	0.7498	0	0	0	0.5000
baseline: hour-only logistic (best-F1 threshold)	0.2718	0.2530	0.9780	0.4020	0.4850

- The model finds 443 of the 545 high validation hours (missing 102) and wrongly flags 162 others.
- F1 stays near 0.77 although AUC is 0.93. Our explanation, which we did not test separately: the label is a sharp cut through a noisy count, so hours just below or above their limit are almost identical.
- On the chronological split the AUC is 0.8125. There 64% of the hours count as 'high' (demand grew in 2012 and the limits come from earlier months), so only the AUC is comparable with the normal split.



Phase 5: ROC and precision-recall curves on validation, final model against the hour-only baseline.

Test predictions

In one sentence: we predict *test.csv* with the unregularised Phase 3 model (R9, 589 features), refit on all labelled rows and floored at 0.

- **Why this model:** nothing beat it. Ridge, Lasso and Elastic Net reach 0.9069, 0.9079 and 0.9069 on validation (against 0.9104), and the Lasso scores 0.8938 against 0.9063 on the chronological split.
- **Why refit on all 10,886 rows (allowed if justified):** the day-grouped 5-fold CV refits the whole recipe in every fold and gives R^2 **0.9148** (RMSE 52.68), and **0.9209** (RMSE 50.75) after flooring. The test days sit between the training days, so 25% more labelled days help.
- **Why floor at 0:** counts cannot be negative. 33 of the 574 test predictions (5.7%) were negative; the lowest was -105.

- **Checks:** the final fit (1,448 iterations) matches the closed-form training loss to 4.7e-6 and is bit-identical on a second run. *sample_submission.csv* has 574 rows in *test.csv* order. The mean prediction (197.4) is close to the training mean (191.6), and every hourly mean prediction is within 25.4 bikes of the hourly training mean.

Pipeline retrospective

phase	model	features	validation (seeded split)	chronological split*
1	linear, own gradient descent	54	R^2 0.6781, RMSE 100.44	R^2 0.6364
2	powers of temp, hum, windspeed, days (degree 3)	62	R^2 0.6858, RMSE 99.24	R^2 0.5099
3	+ hour interactions, no time powers (R9)	589	R^2 0.9104, RMSE 53.00	R^2 0.9063
4	Lasso, $\lambda = 0.2512$	470	R^2 0.9079, RMSE 53.72	R^2 0.8938
5	logistic regression on the Phase 4 features	470	AUC 0.9331, F1 0.7704, accuracy 0.8788	AUC 0.8125

**Phases 1 and 2 were not evaluated chronologically in their own phase; their values are the R0 and R1 rows of the Phase 3 ladder.*

- The big gain came from the **structure** of the model (hour interactions: +0.225 of R^2), not from curves (+0.008) or regularization (-0.0025).
- The time-ordered split exposed the polynomial in days_since_start that random splits hid; out-of-time numbers are always a little lower than random-split ones.
- Every hand-off is hash-checked: Phase 1 into 2, 2 into the Phase 3 ladder, Phase 3's model into 4, Phase 4's 470 features into 5, and Phase 3's recipe into the submission.
- **Limits:** choices made on the seeded validation set (Phase 2 degree, Phase 4 λ, Phase 5 L2 and threshold) make those validation numbers slightly optimistic, as the project rules prescribe; the hum = 0 rows and the 58 missing hours are kept as they are; our explanation of the Phase 5 F1 is untested.